

service/	Business Logic Layer (6 files)
AuthService.java	(Pattern provided)
DonorService.java	
RequestService.java	
MatchService.java	
NotificationService.java	
EmailService.java	
controller/	REST API Controllers (6 files)
AuthController.java	(Pattern provided)
DonorController.java	
RequestController.java	
MatchController.java	
AdminController.java	
NotificationController.java	
security/	Security & JWT (3 files)
JwtTokenProvider.java	(Pattern provided)
JwtAuthenticationFilter.java	
CustomUserDetailsService.java	
exception/	Exception Handling (3 files)
ResourceNotFoundException.java	
UnauthorizedException.java	
GlobalExceptionHandler.java	
util/	Utility Classes
Constants.java	
resources/	
application.properties	Application configuration
application-prod.properties	(Optional: Production config)
test/	Unit & Integration Tests
java/org/lifelink/	
AuthServiceTest.java	
DonorServiceTest.java	
...	(more test files)
frontend/	React Frontend
package.json	NPM dependencies
vite.config.js	Vite build configuration
tailwind.config.js	Tailwind CSS custom theme
postcss.config.js	PostCSS configuration
index.html	HTML entry point
.eslintrc.js	ESLint configuration (optional)
public/	Static Assets
favicon.ico	
logo.png	
assets/	
src/	
main.jsx	React entry point
App.jsx	Main app with routing
index.css	Global styles
pages/	Page Components (14 files)
LandingPage.jsx	Complete landing page
Login.jsx	Login page
Register.jsx	Registration page
About.jsx	About page
Awareness.jsx	Awareness/Education page
NotFound.jsx	404 page
donor/	Donor Module (2 pages)
Dashboard.jsx	(Donor dashboard)

└─ Profile.jsx	(Donor profile managment)
└─ hospital/	🔑 Hospital Module (3 pages)
└─ Dashboard.jsx	(Hospital dashboard)
└─ Requests.jsx	(Manage organ requests)
└─ Search.jsx	(Search compatible donors)
└─ admin/	🔑 Admin Module (4 pages)
└─ Dashboard.jsx	(Admin dashboard with charts)
└─ Donors.jsx	(Verify & manage donors)
└─ Requests.jsx	(Approve organ requests)
└─ Matches.jsx	(Create & manage matches)
└─ components/	🔑 Reusable UI Components (15-20 files)
└─ layout/	
└─ Navbar.jsx	(Top navigation bar)
└─ Sidebar.jsx	(Side navigation)
└─ Footer.jsx	(Footer component)
└─ Layout.jsx	(Main layout wrapper)
└─ common/	
└─ Button.jsx	(Reusable button)
└─ Input.jsx	(Form input)
└─ Card.jsx	(Card component)
└─ Modal.jsx	(Modal dialog)
└─ Table.jsx	(Data table)
└─ Badge.jsx	(Status badge)
└─ Alert.jsx	(Alert messages)
└─ Loader.jsx	(Loading spinner)
└─ forms/	
└─ DonorProfileForm.jsx	(Donor registration form)
└─ OrganRequestForm.jsx	(Create request form)
└─ SearchFilter.jsx	(Search/filter component)
└─ charts/	
└─ StatCard.jsx	(Statistics card)
└─ PieChart.jsx	(Chart.js pie chart)
└─ BarChart.jsx	(Chart.js bar chart)
└─ ProtectedRoute.jsx	(Route guard for auth)
└─ services/	🔑 API Service Layer (6 files)
└─ api.js	(Axios instance with interceptors)
└─ authService.js	(Login, register, logout)
└─ donorService.js	(Donor CRUD operations)
└─ requestService.js	(Request management)
└─ matchService.js	(Matching operations)
└─ notificationService.js	(Notifications)
└─ context/	🔑 React Context (State Management)
└─ AuthContext.jsx	(Authentication state)
└─ hooks/	🔑 Custom React Hooks (Optional)
└─ useAuth.js	(Authentication hook)
└─ useDonor.js	(Donor data hook)
└─ useNotifications.js	(Notifications hook)
└─ utils/	🔑 Utility Functions
└─ helpers.js	(Helper functions)
└─ validators.js	(Form validation)
└─ constants.js	(Constants)
└─ formatters.js	(Date, number formatters)
└─ styles/	🔑 Additional Styles (Optional)
└─ custom.css	(Custom CSS if needed)

File Statistics

Files Already Created (17 files)

- Documentation: 6 files
- Database: 1 file
- Backend: 4 files (config + main class)
- Frontend: 6 files (config + landing page)

Files to Create (~60-70 files)

- Backend Java: 35-40 files
- Frontend React: 25-30 files

Total Lines of Code (when complete)

- Backend: ~15,000 lines
- Frontend: ~8,000 lines
- **Total: ~23,000 lines**

Priority Order for Implementation

Phase 1: Critical Backend (Must have first)

1.  LifeLinkApplication.java (Done)
2. SecurityConfig.java
3. All 8 Entity classes
4. All 8 Repository interfaces
5. AuthService + AuthController
6. JwtTokenProvider + JwtAuthenticationFilter

Phase 2: Core Frontend (Authentication)

1. main.jsx + index.css
2. AuthContext.jsx
3. api.js + authService.js
4. Login.jsx + Register.jsx
5. ProtectedRoute.jsx

Phase 3: Donor Module

1. Backend: DonorService + DonorController
2. Frontend: Donor pages + components

Phase 4: Hospital Module

1. Backend: RequestService + RequestController
2. Frontend: Hospital pages + components

Phase 5: Admin Module

1. Backend: MatchService + AdminController
 2. Frontend: Admin pages + charts
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 Quick Reference

Symbol	Meaning
	Already created (complete)
	To be created (patterns provided)
	Directory/Folder
	Document/File
	Start here / Most important

Legend:

- **Backend patterns** → See [documentation/BACKEND_CODE_REFERENCE.md](#)
- **Setup instructions** → See [documentation/DEPLOYMENT_GUIDE.md](#)
- **API reference** → See [documentation/API_DOCUMENTATION.md](#)